Data.SDR: Similarity, Statistics & Combinatorics Roadmap

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Data.SDR needs three additions to make every match threshold principled: normalized similarity measures, hypergeometric significance tests, and weight-preserving composition operators. This document lists the equations, in order of payoff, beyond the existing set algebra, raw overlap, `kWTA`, boosting, encoders and anomaly score.

Notation. An SDR is a set $x \subseteq \{0, \dots, n-1\}$ with weight $w_x = |x|$.

1. Normalized similarity

Raw overlap $|x \cap y|$ can't be compared across SDRs of different weights, so add the normalized forms: Jaccard, Dice, cosine and Hamming.

$$J(x, y) = \frac{|x \cap y|}{|x \cup y|}, \qquad D(x, y) = \frac{2|x \cap y|}{w_x + w_y}$$

$$\cos(x, y) = \frac{|x \cap y|}{\sqrt{w_x w_y}}, \qquad H(x, y) = |x \triangle y| = w_x + w_y - 2|x \cap y|$$

$1 - J$ and H are true metrics, so you get triangle-inequality pruning in search. Two asymmetric measures matter for HTM-style "is this pattern contained in that context" queries:

$$\text{contain}(x \subseteq y) = \frac{|x \cap y|}{w_x}, \qquad T_{\alpha,\beta}(x, y) = \frac{|x \cap y|}{|x \cap y| + \alpha|x \setminus y| + \beta|y \setminus x|}$$

The second is the Tversky index. $\alpha = \beta = 1$ gives Jaccard, $\alpha = \beta = \frac{1}{2}$ gives Dice, and skewing them gives graded subset queries.

2. Significance: is this overlap meaningful?

This is the most important addition: the Ahmad & Hawkins hypergeometric result. Take a random SDR of weight w drawn from n bits. The probability it overlaps a fixed x in at least θ bits is:

$$P(|x \cap y| \geq \theta) = \sum_{b=\theta}^{\min(w_x, w)} \frac{\binom{w_x}{b}\binom{n-w_x}{w-b}}{\binom{n}{w}}$$

The expected overlap and variance give a z-score for free:

$$\mathbb{E} = \frac{w_x w}{n}, \qquad \mathrm{Var} = \frac{w_x\, w\, (n - w_x)(n - w)}{n^2 (n - 1)}$$

This turns every match threshold θ from a magic number into a chosen false-positive rate. Haskell's `Integer` computes it exactly; use log-gamma for speed at large n.

3. Union capacity

A union of M stored patterns (HTM's temporal-pooling trick) has approximate density:

$$p_M \approx 1 - \left(1 - \frac{w}{n}\right)^M$$

The false-match probability for a fresh SDR tested against that union is a binomial tail:

$$P_{\mathrm{fp}}(M) \approx \sum_{b=\theta}^{w} \binom{w}{b} p_M^{\,b} \,(1 - p_M)^{w-b}$$

Solving $P_{\mathrm{fp}}(M) \leq \varepsilon$ for M tells you how many items a union holds before it lies to you. Export it as `unionCapacity n w θ ε`.

4. Noise and subsampling robustness

Suppose a fraction ρ of x's bits is corrupted, or you store only s of the w bits. With $k = \rho w$ bits lost, the probability the true match still clears θ is again hypergeometric:

$$P(\text{survive}) = \sum_{b=\theta}^{s} \frac{\binom{w-k}{b}\binom{k}{s-b}}{\binom{w}{s}}$$

Together with section 2, this gives the full ROC curve for any choice of (n, w, θ).

5. Combinatorics proper

- **Capacity:** $\binom{n}{w}$ distinct SDRs, carrying $\log_2 \binom{n}{w}$ bits of information.
- **Combinatorial number system (combinadics):** a bijection between SDRs and integers. It gives perfect hashing, compact serialization, exhaustive enumeration, and uniform sampling via `unrank`. Floyd's algorithm is the cheap alternative for sampling alone.

- **Expected overlap of k random SDRs:** useful for designing multi-way coincidence detectors.

Rank, for sorted indices $c_0 < c_1 < \cdots < c_{w-1}$:

$$\text{rank}(x) = \sum_{i=0}^{w-1} \binom{c_i}{i+1} \in \left[0, \binom{n}{w}\right)$$

Expected k-way overlap:

$$\mathbb{E}\,|x_1 \cap \cdots \cap x_k| = n \left(\frac{w}{n}\right)^k$$

6. Composition operators

This is where the "new branch of math" lives. The current algebra ($\cup$, $\cap$, $\setminus$, concat, slice) breaks density: a union of SDRs gets denser, an intersection gets sparser. A closed algebra needs operators that preserve weight.

- **Permutation** ρ (cyclic shift or a fixed random permutation). It preserves weight and similarity, and makes $\rho(x)$ dissimilar to x. A sequence encodes as $\rho^2(a) \cup \rho(b) \cup c$.
- **Context-dependent thinning** (Rachkovskij & Kussul). It restores target density after a union; pick K so the result has weight $\approx w$. The result stays similar to each input, so it is a superposition operator with stable sparsity.
- **Sparse block-code binding** (Laiho et al. 2015; Frady et al. 2021). Partition n into w blocks with one active bit each, then bind per block with modular addition. It is exactly invertible, which gives key–value binding that plain set algebra can't do.

Thinning:

$$\langle x \vee y \rangle = (x \cup y) \cap \bigcup_{k=1}^{K} \rho_k(x \cup y)$$

Block-code binding and its inverse:

$$(x \otimes y)_j = (x_j + y_j) \bmod B, \qquad x = (x \otimes y) \oslash y$$

Laws to QuickCheck:

- $|x \otimes y| = w$
- $J(x \otimes y,\, x) \approx 0$
- $J(\rho x, \rho y) = J(x, y)$
- $J(\langle x \vee y \rangle,\, x) > J_{\text{random}}$

7. Encoder contracts

Each encoder E should state its similarity-vs-distance curve as a law. For a contiguous-block scalar encoder with width w and resolution r:

$$|E(a) \cap E(b)| = \max\left(0,\ w - \left\lceil \frac{|a-b|}{r} \right\rceil\right)$$

Making "overlap is monotone non-increasing in distance" a typeclass law catches bad encoders early.

8. Search at scale

For terabyte-scale in-RAM stores, use MinHash. With a random permutation π, the probability of matching minimum hashes equals the Jaccard similarity:

$$\Pr[\min \pi(x) = \min \pi(y)] = J(x, y)$$

Banded MinHash gives sublinear nearest-neighbor lookup over millions of SDRs. An inverted index from bit to SDR ids is the exact-match complement.

9. Statistics for homeostasis

Per-bit activation frequency f_i and population entropy give boosting a measurable target, where H_b is binary entropy:

$$H_{\text{pop}} = \sum_{i=0}^{n-1} H_b(f_i), \qquad H_b(f) = -f \log_2 f - (1-f)\log_2(1-f)$$

Mutual information between two SDR streams shows whether an encoder or pooler actually carries signal.

Priorities

Implement these three first:

1. Normalized measures plus Tversky (section 1).
2. Hypergeometric significance and union capacity (sections 2–3), which make every threshold in Rhayader principled.
3. Thinning plus block-code binding (section 6), which turns Data.SDR from a set library into an algebra worth a paper.

The proposed home is new `Data.SDR.Similarity` and `Data.SDR.Stats` modules with QuickCheck laws.

References

- Ahmad & Hawkins (2015), *Properties of Sparse Distributed Representations and their Application to Hierarchical Temporal Memory*
- Kanerva, *Sparse Distributed Memory* (1988); *Hyperdimensional Computing* (2009)
- Rachkovskij & Kussul, context-dependent thinning
- Laiho et al. (2015); Frady et al. (2021), sparse block codes and binding